

A bicyclic 30-vertex graph with domination root -4

A counterexample to the proposed 33-vertex minimum

Research note prepared with ChatGPT

18 September 2026

Part II, *Cycle budgets for integer domination roots*, added in batch 67

30 September 2026

Abstract

We construct a connected simple graph on 30 vertices and 31 edges whose domination polynomial has the simple integer root -4 . This answers negatively the explicit question whether 33 is the minimum possible order. The graph is planar, has treewidth two, and has domination number ten. The root certificate reduces to the rational identity

$$21 - \frac{5}{13} + \frac{256}{13} \left(-\frac{1}{2} - \frac{35}{64} \right) = 0.$$

We prove the reduction, give the complete edge list and polynomial, and provide three mutually consistent exact polynomial certificates. A supported-edge construction gives connected planar examples at every order at least 32, and examples with arbitrarily high multiplicity at -4 . The accompanying dependency-free Python programs reproduce the discovery and verify the certificates. No claim is made that 30 is the true minimum order.

[Added 30 September 2026, batch 67.] Part II, from a second manuscript, studies the alternating evaluation $D(G, -1)$. With $\beta(G)$ the cycle rank it proves the sharp bound $1 \leq |D(G, -1)| \leq 3^{\beta(G)}$, a simultaneous divisibility budget for integer roots and their multiplicities, and the exact sets of values of $|D(G, -1)|$: $\{1, 3, 7, 9\}$ for $\beta \leq 2$, with a conventional proof, and a ten-element set for $\beta \leq 3$, computer-assisted. So no graph of cycle rank at most two has the root -6 . It also shows that the zero- C four-cycle search of Part I forces $|D(G, -1)| = 3$, so that search can produce no integer root below -2 other than a simple -4 . Part II constructs no root -6 or -8 .

Main exact result. There is a connected planar graph G_{30} with

$$|V| = 30, \quad |E| = 31, \quad \text{tw}(G_{30}) = 2, \quad D(G_{30}, x) = x^{10}(x + 4)Q_{19}(x),$$

where $Q_{19} \in \mathbb{Z}[x]$ and $Q_{19}(-4) = -38\,318\,272 \neq 0$. The graph, not merely an abstract existence argument, is specified below.

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Part I. A bicyclic 30-vertex graph with domination root -4

1 The question selected and the scope of the answer

For a finite simple graph $G = (V, E)$, a set $S \subseteq V$ is *dominating* when every vertex outside S has a neighbor in S . The domination polynomial is the generating polynomial

$$D(G, x) = \sum_{\substack{S \subseteq V \\ S \text{ dominating}}} x^{|S|} = \sum_{k=0}^{|V|} d(G, k) x^k. \quad (1.1)$$

A domination root is a zero of this polynomial. The coefficients count sets; all evaluations and cancellations in this note are exact.

Alikhani and Griswold [1] gave a 33-vertex graph with domination root -4 , already disproving the earlier conjecture that the only integer domination roots are 0 and -2 . Their Section 4, question 2, asks whether 33 is minimal. We attack that follow-up question, not the already-refuted original conjecture. The present construction shows that 33 is *not* minimal. It also reduces the cyclomatic number from four in their example to two.

The result is an explicit finite certificate with a conventional proof. It is not a claim of a globally smallest graph, a classification of integer domination roots, or a resolution of whether -6 occurs. Literature searches through the date of this note did not locate the same improvement, but cannot establish publication priority. Independent human review remains appropriate before making a novelty claim in a submission.

[Added 30 September 2026, batch 67] Part II and the root -6 . Whether -6 occurs is still open, but Part II narrows it. Corollary 16.2 shows that no graph of cycle rank at most two has the root -6 , and Corollary 17.2 that a root -6 of multiplicity at least two needs cycle rank at least four. The eight-vertex graph X_5 of Section 17.3 shows that the divisibility obstruction alone cannot exclude -6 at cycle rank three. Neither part constructs a graph with root -6 or -8 .

2 The graph and its elementary invariants

Let $V(G_{30}) = \{0, 1, \dots, 29\}$. Its edge set is the following set of unordered pairs:

$$\begin{aligned}
 E(G_{30}) = \{ & \{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 0\}, \{1, 4\}, \{2, 4\}, \\
 & \{4, 5\}, \{4, 6\}, \{4, 7\}, \{4, 8\}, \{4, 9\}, \{1, 10\}, \{10, 11\}, \\
 & \{2, 12\}, \{12, 13\}, \{12, 14\}, \{14, 15\}, \{15, 16\}, \\
 & \{14, 17\}, \{17, 18\}, \{14, 19\}, \{19, 20\}, \{14, 21\}, \\
 & \{21, 22\}, \{21, 23\}, \{21, 24\}, \{3, 25\}, \{25, 26\}, \\
 & \{25, 27\}, \{27, 28\}, \{28, 29\} \}.
 \end{aligned}
 \tag{2.1}$$

For machine use, `data/graph30.json` and `data/edges.txt` contain the same edge records as pairs of integers.

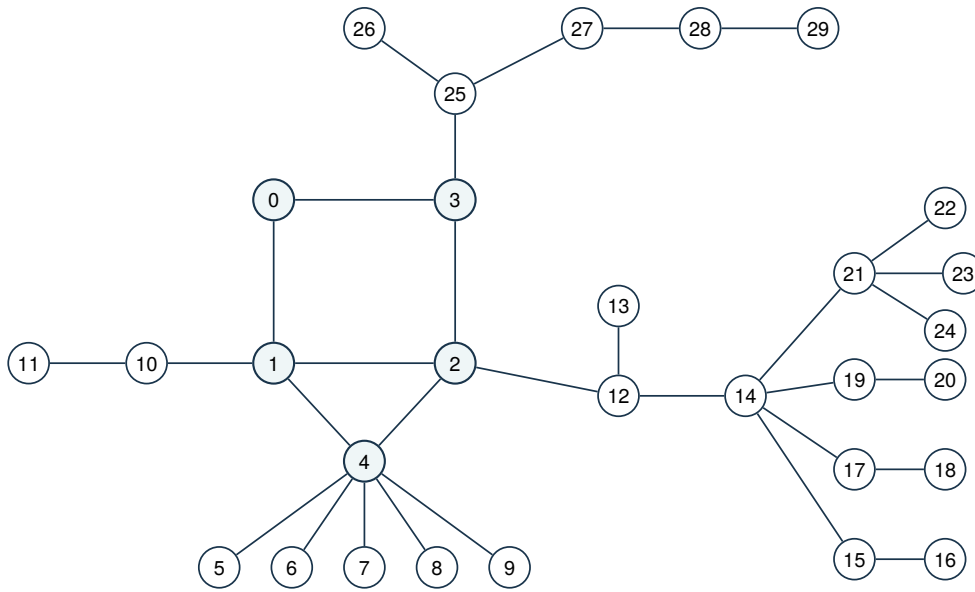


Figure 1: The exact 30-vertex graph. Shaded vertices form its five-vertex 2-core. The displayed drawing is planar.

An equivalent structural description starts with the cycle $0 - 1 - 2 - 3 - 0$. Add a common neighbor 4 of vertices 1, 2, and give 4 five leaves. Attach to vertex 1 a two-vertex rooted path. Attach to vertex 2 the rooted tree on vertices $12, \dots, 24$. Attach to vertex 3 the rooted tree on vertices $25, \dots, 29$. Only the root of each attached tree meets the cycle, by one added edge. Vertex 0 has no attached tree.

Proposition 2.1. *The graph G_{30} is connected and planar, has 30 vertices, 31 edges, 15 leaves, maximum degree seven, cyclomatic number two, treewidth two, and domination number ten.*

Proof. The construction gives connectivity and the counts directly. Removing all pendant trees leaves the cycle together with the vertex 4 and its two edges to 1, 2. This core is planar; attaching trees preserves planarity. For a connected graph the cyclomatic number is $|E| - |V| + 1$, which here is $31 - 30 + 1 = 2$.

For completeness, a tree decomposition of the core has the three bags

$$\{1, 2, 4\}, \quad \{0, 1, 2\}, \quad \{0, 2, 3\},$$

joined in the displayed order. Every core edge lies in a bag, and the bags containing any fixed vertex form a connected subtree. Attach one bag $\{u, v\}$ for each edge in a pendant tree, extending from the bag

containing its parent vertex. The largest bag has size three, so the width is at most two. The graph contains a cycle and is not a forest, so its treewidth is not at most one.

To establish the domination number without the polynomial calculation, consider the nine pairwise disjoint closed neighborhoods of leaves

$$\{4, 5\}, \{10, 11\}, \{12, 13\}, \{15, 16\}, \{17, 18\}, \{19, 20\}, \{21, 22\}, \{25, 26\}, \{28, 29\}.$$

Every dominating set must meet each one. It must also meet $N[0] = \{0, 1, 3\}$, disjoint from all nine. Thus its size is at least ten. The set

$$\{0, 4, 10, 12, 15, 17, 19, 21, 25, 28\}$$

dominates every vertex, proving the matching upper bound. $\square$

The cyclomatic number has an additional interpretation. Oboudi's Theorem 7 [2] excludes integer roots other than 0, -2 for trees and unicyclic graphs. Combining that published result with the construction below gives

$$\min\{|E(G)| - |V(G)| + 1 : G \text{ connected}, D(G, -4) = 0\} = 2. \quad (2.2)$$

The upper bound in (2.2) is proved here; its lower bound uses the cited theorem. This is an exact minimum of *cycle rank*, not an assertion that 30 is a minimum of *order*.

[Added 30 September 2026, batch 67] Cycle rank in Part II. Part II writes $\beta(G) = |E(G)| - |V(G)| + c(G)$, with $c(G)$ the number of components; for connected graphs it is the cyclomatic number used here. Part II does not re-prove the lower bound in (2.2): its obstruction at -1 does not exclude a root -4 at cycle rank one ($3 \in \mathcal{A}_1$ in Theorem 16.1), so that bound still rests on Oboudi's theorem. Part II's Theorem 13.1 adds that a root -4 of multiplicity r needs cycle rank at least r .

3 Rooted-tree states: the local calculation

The three-state recurrence below is elementary. We include the details so that the root certificate does not rely on any black-box graph polynomial implementation.

Definition 3.1. For a tree T with distinguished root v , let $A_T, B_T, C_T \in \mathbb{Z}[x]$ be the weighted sums of subsets of $V(T)$ with these respective properties:

- (a) every vertex is dominated and v is selected;
- (b) every vertex is dominated and v is not selected;
- (c) every vertex except v is dominated, and v is not dominated.

The weight of a subset S is $x^{|S|}$. In state C_T , the root is necessarily unselected. Set $P_T = A_T + B_T = D(T, x)$.

Write $\mathcal{R}(T_1, \dots, T_s)$ for the rooted tree obtained by adjoining one new root and joining it to the roots of disjoint copies of $T_1, \dots, T_s$. In particular, $L = \mathcal{R}()$ is one vertex.

Lemma 3.2 (Rooted recurrence). *For $T = \mathcal{R}(T_1, \dots, T_s)$,*

$$A_T = x \prod_{j=1}^s (A_{T_j} + B_{T_j} + C_{T_j}), \quad (3.1)$$

$$B_T = \prod_{j=1}^s (A_{T_j} + B_{T_j}) - \prod_{j=1}^s B_{T_j}, \quad (3.2)$$

$$C_T = \prod_{j=1}^s B_{T_j}. \quad (3.3)$$

Empty products are one, so $(A_L, B_L, C_L) = (x, 0, 1)$.

Proof. If the new root is selected, it contributes x , and each child root may be in any of the three states: a formerly undominated child root is now dominated by its selected parent. This proves (3.1). If the new root is unselected, each child subtree must internally dominate its own root, allowing only A and B . Among these choices, precisely those in which every child is in state B leave the new root undominated. Subtracting that class gives (3.2), and retaining it gives (3.3). All decompositions preserve the number of selected vertices, proving the polynomial identities. $\square$

If a rooted tree has a leaf immediately adjacent to its root, then $C_T = 0$ as a polynomial: such a leaf contributes the factor $B_L = 0$ to (3.3). Combinatorially, an undominated root would force its private leaf to be both unselected and undominated.

For our graph define

$$\begin{aligned} E &= \mathcal{R}(L), & J &= \mathcal{R}(E), & S_3 &= \mathcal{R}(L, L, L), \\ W &= \mathcal{R}(E, E, E, S_3), & T &= \mathcal{R}(L, W), & K &= \mathcal{R}(L, J), \\ S_5 &= \mathcal{R}(L, L, L, L, L). \end{aligned}$$

The four external gadgets are E, T, K, S_5 , of orders 2, 13, 5, 6. Their roots are respectively the vertices 10, 12, 25, 4. The first three attach to one cycle vertex; the last attaches to both 1, 2. Each of the four has $C = 0$ identically.

Tree	Order	$A(-4)$	$B(-4)$	$C(-4)$	$B/(A+B)$ at -4
L	1	-4	0	1	—
E	2	12	-4	0	-1/2
J	3	-32	12	-4	—
S_3	4	108	-64	0	-16/11
W	11	-90 112	18 432	4 096	—
T	13	-811 008	286 720	0	-35/64
K	5	-288	80	0	-5/13
S_5	6	972	-1 024	0	256/13

Table 1: Exact rooted states. Dashes suppress ratios not used as zero- C attachments. Every entry follows from Lemma 3.2.

Here are the larger arithmetic steps, to make the table checkable by hand. For E , $A + B = 8$; for S_3 , $A + B = 44$. Thus

$$\begin{aligned} A_W &= -4 \cdot 8^3 \cdot 44 = -90\,112, \\ B_W &= 8^3 \cdot 44 - (-4)^3(-64) = 18\,432, & C_W &= 4\,096. \end{aligned}$$

Therefore $A_W + B_W + C_W = -67\,584$, while $A_W + B_W = -71\,680$. Adding the new root and its leaf gives

$$\begin{aligned} A_T &= -4(-3)(-67\,584) = -811\,008, \\ B_T &= (-4)(-71\,680) = 286\,720, & C_T &= 0. \end{aligned}$$

Likewise $(A_J, B_J, C_J) = (-32, 12, -4)$, and hence $A_K = -4(-3)(-24) = -288$, $B_K = (-4)(-20) = 80$. Finally $A_{S_5} = -4(-3)^5 = 972$ and $B_{S_5} = (-4)^5 = -1024$.

4 A four-cycle identity and the root proof

We now prove the identity that turns the table into a root certificate. Although the search used weighted states at one value of x , all four gadgets in the final graph satisfy the stronger polynomial condition $C = 0$.

Proposition 4.1 (One hub and one undecorated corner). *Start with C_4 on 0, 1, 2, 3. Attach zero- C rooted gadgets F_1, F_2, F_3 to vertices 1, 2, 3, respectively, and attach a zero- C rooted gadget H by joining its root to both 1, 2. No gadget is attached to 0. Write*

$$P_i = A_i + B_i, \quad P_H = A_H + B_H, \quad a = \frac{B_1}{P_1}, \quad b = \frac{B_2}{P_2}, \quad c = \frac{B_3}{P_3}, \quad r = \frac{B_H}{P_H}, \quad \mathcal{P} = P_1 P_2 P_3 P_H.$$

As an identity of rational functions (and after clearing denominators, of polynomials), the resulting graph G satisfies

$$D(G, x) = x\mathcal{P}[x^3 + 4x^2 + 6x + 3 - c - r(a + b)]. \quad (4.1)$$

Proof. First sum over configurations that dominate all gadget vertices but impose no domination requirement on the four cycle vertices. A root cannot be in state C , so its total internal weight is P , independently of which adjacent cycle vertices are selected. The unrestricted total is consequently $\mathcal{P}(1+x)^4$.

Apply inclusion–exclusion to the events that specified cycle vertices are undominated. For $U \subseteq \{0, 1, 2, 3\}$, this forces every cycle vertex in $N_{C_4}[U]$ to be unselected. The remaining cycle vertices contribute $(1+x)^{4-|N_{C_4}[U]|}$. Every attached root meeting a vertex of U must also be unselected, replacing its factor P by B . In particular, the shared root contributes r once whenever $U \cap \{1, 2\} \neq \emptyset$, not twice. With $u_0 = 1, u_1 = a, u_2 = b, u_3 = c$, this gives

$$\frac{D(G, x)}{\mathcal{P}} = \sum_{U \subseteq \{0, 1, 2, 3\}} (-1)^{|U|} (1+x)^{4-|N_{C_4}[U]|} \prod_{i \in U} u_i r^{\mathbf{1}_{U \cap \{1, 2\} \neq \emptyset}}. \quad (4.2)$$

Put $t = 1 + x$. The empty-set contribution is t^4 , and singleton contributions sum to $-t[1 + c + r(a + b)]$. Any two distinct cycle vertices have closed-neighborhood union equal to the entire cycle. The contributions by cardinality two, three and four are therefore

$$\begin{aligned} & c + r(a + b + ab + ac + bc), \\ & -r(ab + ac + bc + abc), \\ & rabc, \end{aligned}$$

whose sum is $c + r(a + b)$. Thus (4.2) becomes

$$t^4 - t + (1 - t)[c + r(a + b)] = x[x^3 + 4x^2 + 6x + 3 - c - r(a + b)],$$

proving the assertion. Clearing denominators also proves the identity at values where a displayed ratio is undefined. $\square$

Theorem 4.2 (The counterexample). *The graph specified by (2.1) satisfies $D(G_{30}, -4) = 0$. Consequently, the answer to the 33-vertex minimality question is no.*

Proof. Apply Proposition 4.1 with $F_1 = E, F_2 = T, F_3 = K, H = S_5$. At $x = -4$,

$$(P_1, P_2, P_3, P_H) = (8, -524\,288, -208, -52), \quad \mathcal{P} = -45\,365\,592\,064 \neq 0.$$

Table 1 gives

$$a = -\frac{1}{2}, \quad b = -\frac{35}{64}, \quad c = -\frac{5}{13}, \quad r = \frac{256}{13}.$$

At -4 , the bracket in (4.1) is $-21 - c - r(a + b)$. But

$$a + b = -\frac{67}{64}, \quad r(a + b) = -\frac{268}{13}, \quad 21 + c = \frac{268}{13}.$$

The bracket is exactly zero. Equivalently,

$$\frac{D(G_{30}, -4)}{4\mathcal{P}} = 21 - \frac{5}{13} + \frac{256}{13} \left(-\frac{1}{2} - \frac{35}{64} \right) = 0. \quad (4.3)$$

This is an equality in $\mathbb{Q}$, not a numerical approximation. $\square$

Remark 4.3. The proof uses only the explicit graph and the elementary state and inclusion–exclusion identities proved above. It does not assume the correctness of a numerical factorization, any search program, or any particular normalization of the transfer matrices in earlier work.

5 The full polynomial and independent certificates

The nonzero coefficients are listed in Table 2. All coefficients below degree ten vanish. The last coefficient is one, as required because the whole vertex set is the unique set of size 30.

k	$d(G, k)$	k	$d(G, k)$	k	$d(G, k)$
10	256	17	4 526 996	24	342 684
11	4 192	18	5 803 019	25	99 255
12	32 576	19	6 049 225	26	22 114
13	159 418	20	5 150 665	27	3 651
14	550 047	21	3 583 938	28	420
15	1 420 449	22	2 031 824	29	30
16	2 845 950	23	932 405	30	1

Table 2: The complete nonzero coefficient list of $D(G_{30}, x)$.

Exact division gives

$$D(G_{30}, x) = x^{10}(x + 4)Q_{19}(x), \quad (5.1)$$

where

$$\begin{aligned} Q_{19}(x) = & x^{19} + 26x^{18} + 316x^{17} + 2387x^{16} + 12566x^{15} \\ & + 48991x^{14} + 146720x^{13} + 345525x^{12} + 649724x^{11} \\ & + 985042x^{10} + 1210497x^9 + 1207237x^8 + 974071x^7 \\ & + 630712x^6 + 323102x^5 + 128041x^4 + 37883x^3 \\ & + 7886x^2 + 1032x + 64. \end{aligned} \quad (5.2)$$

No irreducibility assertion is needed. Substitution yields

$$Q_{19}(-4) = -38\,318\,272, \quad D'(G_{30}, -4) = (-4)^{10}Q_{19}(-4) = -40\,179\,620\,380\,672 \neq 0. \quad (5.3)$$

This proves that -4 is a simple root.

[Added 30 September 2026, batch 67] A second route to simplicity. Part II computes $D(G_{30}, -1) = -3$ by its feedback-edge algorithm (Section 19.2). A double root -4 would make 3^2 divide $D(G_{30}, -1)$ (Lemma 10.2), so the root is simple without the derivative in (5.3). The root itself still rests on the certificates of this section.

5.1 Certificate A: expansion of the four-cycle identity

Proposition 4.1 has the denominator-free form

$$\begin{aligned} D(G, x) = & x[(x^3 + 4x^2 + 6x + 3)P_1P_2P_3P_H \\ & - B_HP_3(B_1P_2 + B_2P_1) - B_3P_1P_2P_H]. \end{aligned} \quad (5.4)$$

The following compact gadget polynomials, together with (5.4), are one symbolic certificate for every coefficient. Let

$$p = x(x + 2), \quad s = x(1 + x)^3 + x^3.$$

Then

$$\begin{aligned} P_1 &= p, & B_1 &= x, \\ P_H &= x(1 + x)^5 + x^5, & B_H &= x^5, \\ P_3 &= x^2(x^3 + 5x^2 + 8x + 3), & B_3 &= x^2(x^2 + 3x + 1), \\ P_2 &= x(x + 1)(x + 2)p^3s - x^7, & B_2 &= x(x + 1)p^3s - x^7. \end{aligned} \quad (5.5)$$

These follow directly from Lemma 3.2. Expanding (5.4) with (5.5) gives Table 2 and (5.1).

5.2 Certificate B: stripping trees and enumerating the 2-core

Remove vertices of current degree zero or one successively. At a removed vertex, compute its three rooted polynomial states and absorb them into its surviving neighbor. The remaining 2-core here has five vertices: 0, 1, 2, 3, 4.

At each remaining core vertex v , let

$$T_v = \prod_j (A_j + B_j + C_j), \quad U_v = \prod_j (A_j + B_j), \quad V_v = \prod_j B_j,$$

where the product runs over its absorbed child trees. For a selected core subset S , the local contribution of v is

$$\begin{cases} xT_v, & v \in S, \\ U_v, & v \notin S \text{ and } N_{\text{core}}(v) \cap S \neq \emptyset, \\ U_v - V_v, & v \notin S \text{ and } N_{\text{core}}(v) \cap S = \emptyset. \end{cases} \quad (5.6)$$

Multiplying local factors and summing over all $2^5 = 32$ subsets gives the full polynomial. Formula (5.6) is exactly the distinction between root selected, root dominated by a core neighbor, and root needing domination from a child. Thus it counts each dominating set once. This calculation does not use the special four-cycle identity.

5.3 Certificate C: an independent enumeration over non-leaves

The following identity avoids rooted states and 2-core stripping altogether.

Proposition 5.1 (Leaf-subset certificate). *Let G be a finite graph with no component isomorphic to K_2 . Let L be its set of degree-one vertices and $C = V \setminus L$. For $v \in C$, set $t_v = |N(v) \cap L|$. Call $S \subseteq C$ admissible when every $v \in C$ with $t_v = 0$ is dominated by S in $G[C]$. Then*

$$D(G, x) = \sum_{S \text{ admissible}} x^{|S| + \sum_{v \in C \setminus S} t_v} (1 + x)^{\sum_{v \in S} t_v}. \quad (5.7)$$

Proof. Fix the selected non-leaves S . If a non-leaf v is selected, its adjacent leaves may each be selected or not, independently, giving $(1 + x)^{t_v}$. If v is not selected, every one of those leaves must be selected, contributing x^{t_v} . When $t_v > 0$, these forced leaves also dominate v . When $t_v = 0$, the only way to dominate an unselected v is through the selected non-leaves; this is precisely admissibility. The exclusion of isolated-edge components ensures that every leaf has a non-leaf neighbor. These disjoint cases exhaust all dominating sets and prove (5.7). $\square$

For G_{30} , $|L| = |C| = 15$, so the independent computation tests $2^{15} = 32768$ non-leaf subsets. Exactly 23065 are admissible. Grouping the summands in (5.7) by their two exponents gives 118 terms $m_{a,b}x^a(1+x)^b$. The file `data/leaf_certificate.csv` records all triples $(a, b, m_{a,b})$. Binomial expansion reproduces Table 2, and direct substitution in that 118-term sum gives zero at -4 .

6 Connected families and arbitrary multiplicity

There is a simple way to preserve the root without leaving connected graphs. The underlying supported-edge factorization is elementary and related to the pendant-star products in Oboudi [2]; it is not claimed as a new general principle here.

Call a vertex a *support vertex* when it has a neighbor of degree one, called a private leaf.

Lemma 6.1 (Edges between support vertices). *Suppose distinct vertices u, v each have a private leaf outside $\{u, v\}$, and these leaves remain private after the edge uv is added or deleted. Toggling uv does not change the family of dominating vertex subsets, hence does not change the domination polynomial.*

Proof. A dominating set must select either a private leaf of u or u itself, because it must dominate that leaf. Either choice already dominates u , independently of the edge uv . The same holds for v . Every other vertex has exactly the same neighbors before and after the change. Thus a vertex subset dominates one graph exactly when it dominates the other. $\square$

For disjoint graphs, independently choosing dominating sets proves

$$D(G \dot{\cup} H, x) = D(G, x)D(H, x). \quad (6.1)$$

Combining this with the lemma connects factors without changing the product.

Theorem 6.2 (A bounded-width connected family). *For each integer $t \geq 0$, there is a connected planar graph G_t with $30 + 2t$ vertices, cyclomatic number two, treewidth two, maximum degree seven, and*

$$D(G_t, x) = [x(x+2)]^t D(G_{30}, x). \quad (6.2)$$

In particular, -4 is a simple root of every $D(G_t, x)$.

Proof. For $t = 0$, use G_{30} . For $t \geq 1$, take G_{30} and t disjoint edges $a_i b_i$. Retain each b_i as a private leaf. Join vertex 10 to a_1 , and join a_i to a_{i+1} for $1 \leq i < t$. Vertex 10 retains its leaf 11. Each added connecting edge joins support vertices and therefore changes no dominating sets. Since $D(K_2, x) = 2x + x^2 = x(x+2)$, equations (6.1) and (6.2) follow.

The new part is a pendant tree, so planarity, treewidth two, and cycle rank two are preserved. There are $2t$ new vertices and $2t$ new edges. Vertex 10 acquires at most one new incident edge; every a_i has degree at most three. The old degree-seven vertex 4 remains, so the maximum degree is still seven. Finally $(-4)(-4+2) = 8 \neq 0$, so multiplication by $[x(x+2)]^t$ does not alter the multiplicity at -4 . $\square$

Corollary 6.3 (All sufficiently large orders). *There is a connected planar graph of treewidth two and cycle rank two with domination root -4 at order 30 and at every order $n \geq 32$.*

Proof. Besides attaching an isolated edge as above, we may attach a disjoint star $K_{1,2}$ by its center to a support vertex. Lemma 6.1 gives an additional factor

$$D(K_{1,2}, x) = x(1+x)^2 + x^2 = x(x^2 + 3x + 1).$$

At -4 this factor is $-20 \neq 0$; it adds three vertices and a pendant tree. Every integer $n - 30 \geq 2$ is a nonnegative integer combination of two and three. Using one three-vertex attachment for odd $n - 30$, and otherwise only two-vertex attachments, proves the claim. This construction asserts nothing about order 31. $\square$

Theorem 6.4 (Unbounded root multiplicity in connected graphs). *For every integer $k \geq 1$, there is a connected planar graph H_k of order $30k$, treewidth two, and maximum degree seven in which -4 has multiplicity exactly k .*

Proof. Take k disjoint copies of G_{30} , and connect their copies of vertex 10 in a path. Every connecting edge joins support vertices retaining private leaves. Thus

$$D(H_k, x) = D(G_{30}, x)^k.$$

The root has multiplicity exactly k by (5.3). The connecting edges are bridges, so they preserve planarity and do not increase treewidth beyond two. Internal copies of vertex 10 have degree four after the operation; the maximum degree is still seven. $\square$

These results strengthen the bare infinite-family question in [1, Section 4, question 3]: disconnected examples would follow immediately from disjoint union, whereas the constructions here remain connected and structurally sparse.

[Added 30 September 2026, batch 67] Cycle cost of multiplicity. The graph H_k of Theorem 6.4 has cycle rank $2k$ (each copy of G_{30} brings two independent cycles and the connecting edges are bridges). Part II's Theorem 13.1 shows that multiplicity k at -4 needs cycle rank at least k , and its Question 21.3 asks for the true minimum; multiplicity two at cycle rank two would need $|D(G, -1)| = 9$. Part I's H_k is unrelated to Part II's eight-vertex graph X_5 .

7 How the search was made finite and reproducible

The successful search was not a uniform enumeration of 30-vertex graphs. It used the sparse core to turn a combinatorial search into rational arithmetic on a catalogue of small rooted trees.

7.1 The catalogue and the exact target equation

A rooted tree is stored as the canonically ordered tuple of the shapes of its children. The empty tuple is a leaf. Recursively enumerate multisets of previously generated trees whose total order is $n - 1$, then place a new root above them. This generates each unlabeled rooted tree of order n exactly once: deleting the root yields a uniquely determined multiset of rooted child components.

For each tree compute $(A(-4), B(-4), C(-4))$ by Lemma 3.2. Retain its rational ratio

$$q = \frac{B(-4)}{A(-4) + B(-4)} \quad \text{when} \quad C(-4) = 0, \quad A(-4) + B(-4) \neq 0.$$

The condition $C(-4) = 0$ is sufficient for a weighted search at the single point -4 ; it need not hold identically for every candidate. All final gadgets happen to satisfy $C(x) = 0$ identically, which makes the polynomial proof particularly clean.

At a cycle site, disjoint branches give a product of their ratios. Store a least-order realization for every attainable product, including the empty forest with ratio one and order zero. Discarding larger realizations of the same ratio is valid for this search: they have identical normalized effects at -4 , and cannot improve its order objective. The catalogue contains all such minimum-cost products within the prescribed forest-size budget.

For a shared star hub with $m \geq 1$ leaves,

$$r_m = \frac{(-4)^m}{-4(-3)^m + (-4)^m} = \frac{4^{m-1}}{4^{m-1} - 3^m}. \quad (7.1)$$

The denominator is nonzero, since a power of four cannot equal a positive power of three. Leaving vertex 0 undecorated reduces the root search, by Proposition 4.1, to

$$c = -21 - r_m(a + b). \quad (7.2)$$

Thus enumerate two stored ratios a, b , compute the required c , and look it up exactly. No tolerance or root approximation occurs. If the three forests have sizes n_a, n_b, n_c , the total order is

$$n = 5 + m + n_a + n_b + n_c. \quad (7.3)$$

The hit has $(m, n_a, n_b, n_c) = (5, 2, 13, 5)$, giving $n = 30$.

7.2 The supplied run and its limitations

The portable reconstruction program uses rooted trees and site forests of order at most 14 and hub sizes $1 \leq m \leq 15$. It starts with the strict target $n < 33$, tries the five-leaf hub first, and then tests the other permitted hub sizes. It retains improving hits and checks the graph of every hit with the separate 2-core polynomial routine. A representative completed run gives:

Quantity	Recorded value
Distinct single-tree ratios	4 198
Distinct forest ratios, including the empty forest	6 118
Exact rational candidate lookups	2 547 880
Best order returned	30
Time in the execution environment	6.154 seconds

The time is an observation on this environment, not a complexity bound or a cross-machine promise. The deterministic counts and the hit are stored in `results/search.json`. The number of rooted trees at each order from 1 through 14 is also recorded there.

In this bounded search, no smaller realization was returned. That is *not* an exhaustive lower bound for all graphs, all bicyclic graphs, or even all rooted attachments to this core: an individual site forest was limited to 14 vertices. Searching other cores, allowing a decorated fourth corner, or allowing larger individual forests can escape the tested class. The abstract minimum remains open in this note.

[Added 30 September 2026, batch 67] What escapes the tested class. For the *order* of a root -4 the last sentence stands. For any *other* integer root it does not: by Part II's Theorem 19.1, every graph built on this core from rooted trees with $C = 0$ at the target, with any number of trees of any size at each site, a hub carrying at least one pendant leaf, and a decorated fourth corner allowed, has $|D(G, -1)| = 3$. Such a graph has no integer root below -2 except a simple -4 . A search for another root has to leave this architecture altogether, for example by changing the core or by admitting trees with $C \neq 0$ at the target.

7.3 Verification and computational trust

The complete coefficient list was recomputed by all three routes in Section 5: the symbolic four-cycle certificate, the 2-core state algorithm, and the independent leaf-subset identity. The latter two are genuinely different enumerations, although all are implemented in the same Python runtime. They are not independent proof-assistant formalizations.

The two general graph algorithms were also compared against direct enumeration of all vertex subsets for *every labeled simple graph on zero through five vertices*: 1100 graphs in total. Further checks used 120 deterministically seeded graphs on six through nine vertices. Five instances of the connected extension in Theorem 6.2 were checked as polynomial identities. All tests passed.

The verification run recorded about 0.23 seconds. The important properties are the exact outputs, not the timing. All polynomial arithmetic is integer arithmetic, and all search ratios use Python's `Fraction`. No symbolic algebra package, network service, cached pickle, numerical optimizer, or proprietary system is required to run the supplied code.

8 Conclusions and remaining targets

The attack gives a concrete answer to the selected question: a graph with domination root -4 need not have 33 vertices. The explicit order-30 example also demonstrates that cycle rank two and treewidth two already suffice. The cancellation is additive in three rooted ratios and a single shared-star ratio; it does not require a large or highly connected core.

The demonstrated improvement is an *upper bound*: the minimum order is at most 30. A serious lower-bound project should canonicalize possible small cores and carry exact rooted-state catalogues with a global vertex budget, rather than infer minimality from an unsuccessful restricted search. An especially natural next target is whether some graph on at most 29 vertices has root -4 .

For other integer targets z , the same core identity yields

$$c + r(a + b) = z^3 + 4z^2 + 6z + 3,$$

with the rooted states evaluated at z . This provides a concrete search equation, but this note establishes no graph with root -6 or -8 . Nor does it make a claim that all negative even integers occur. The value of the present result is the fully specified, independently checked counterexample and its complete elementary proof.

[Added 30 September 2026, batch 67] The search equation for other targets is answered negatively. Part II settles what this search equation can give. Take an integer $z = -k$ with $k \geq 3$, site trees with $C(z) = 0$ (any number at every corner, the fourth corner included) and a star hub with at least one leaf, as in (7.1). Theorem 19.1 shows that every such graph has $|D(G, -1)| = 3$. Since $k - 1$ must divide $D(G, -1)$ at a root $-k$ (Lemma 10.2), the equation has a solution only at $z = -4$, and then the root is simple. The argument that the proof applies to the site trees applies equally to a hub tree with $C(z) = 0$, as Proposition 4.1 allows: such a hub, too, is never selected and is already dominated at $x = -1$, so it can be deleted. (This extension is the present write's; the manuscript states only the star hub.) So the search equation yields no root -6 or -8 . Whether such roots exist is a different question, and it stays open. Part II excludes -6 for every graph of cycle rank at most two (Corollary 16.2) and lists, in Section 21, the cases at which a search can still aim.

Part II. Cycle budgets for integer domination roots

9 Research context, contribution, and status

9.1 Provenance of Part II

Part II prints batch-67 manuscript 04 of ProveIt's incoming reports, *Cycle Budgets for Integer Domination Roots: A sharp alternating-sum bound, exact low-cycle spectra, and a structural obstruction to a repository search method*, dated 30 September 2026. Its title page reads “Prepared with ChatGPT for Vladimir Reshetnikov”. It arrived as `ProveIt_Domination_Cycle_Obstructions.zip` in commit `ffddaa8b9` and was placed in `d281a285a`. It was written against ProveIt commit `0744012ba29db0e345c44be953c6de3e000439e6`. It read Part I's `README.md`, `domination_root_30.tex` and `code/search.py`. No commit changed this report between that pin and the placement, so everything the manuscript says about “the repository report” refers to Part I as printed here. Part II is a single manuscript; nothing was merged from elsewhere.

Every result, proof, example, remark, question and limitation of the manuscript is printed. The changes are these. Its citations of the repository report are now references to Part I. Its labels carry the prefix `cb:`. Three symbols are renamed (Section 9.2). Its multiplicities m_j in the abstract are written r_j , as in its own theorems. Its bibliography is merged with Part I's. Its delivered programs and data are shipped unchanged under the prefix `02-cycle-budgets-` (the report's README lists them and explains how to rerun them). Text added in this write is marked “[Added 30 September 2026, batch 67]”.

The manuscript's own summary follows. With $\beta(G) = |E(G)| - |V(G)| + c(G)$ the cycle rank, it proves the sharp bound $1 \leq |D(G, -1)| \leq 3^{\beta(G)}$, with equality at the upper end for connected planar subcubic graphs of every positive cycle rank, using a finite signed-state lemma for partially constrained forests and a three-way partition for every feedback edge. Consequently integer domination roots satisfy the simultaneous multiplicity budget $\prod_j (k_j - 1)^{r_j} \mid |D(G, -1)|$ and $\prod_j (k_j - 1)^{r_j} \leq 3^{\beta(G)}$ whenever the distinct roots $-k_j$, $k_j \geq 2$, have multiplicities r_j . The possible absolute evaluations for cycle rank at most two and three are exactly $\{1, 3, 7, 9\}$ and $\{1, 3, 5, 7, 9, 11, 15, 17, 21, 27\}$. The two-cycle classification has an explicit matrix proof. The three-cycle classification reduces to fifteen multigraph kernels and 26,688 exact residue assignments, verified by a dependency-free program. Thus a root -6 requires at least three independent cycles, and a root -14 at least four. A separate theorem shows that the zero- C four-cycle search architecture of Part I cannot produce any integer root below -2 other than -4 , irrespective of branch sizes. These are structural restrictions, not constructions of roots -6 or -8 . No global classification of integer domination roots or publication-priority claim is made.

Principal resolved question (Part II). How much cycle complexity is required by integer roots and their multiplicities? Every such graph obeys

$$\sum_j r_j \log_3(k_j - 1) \leq \beta(G).$$

The exact low-cycle spectra strengthen this numerical bound: in particular, $D(G, -6) \neq 0$ whenever $\beta(G) \leq 2$.

9.2 Conventions: symbols shared with Part I

Part II keeps the manuscript's notation, with three renamings. Inside Part II the letters in the left column of the table below have the Part II meaning; the right column gives the Part I meaning that a reader might wrongly carry over.

Symbol	In Part II	In Part I (do not confuse)
$M_1, \dots, M_4, M_\ell$	3×3 path transfer matrices, (15.1)–(15.4); the manuscript writes F_ℓ	F_1, F_2, F_3 : the rooted gadgets at the corners 1, 2, 3 in Proposition 4.1
X_5	the eight-vertex tricyclic graph with $D(X_5, -1) = 5$, (17.2); the manuscript writes H_5	H_5 : five copies of G_{30} in Theorem 6.4, a 150-vertex graph
G_b^{ch}	the subcubic chain of b four-cycles (sharpness in Theorem 12.1); the manuscript writes G_b	G_t : the family of Theorem 6.2
W	the matrix $\text{diag}(1, -1, -1)$	the rooted tree $\mathcal{R}(E, E, E, S_3)$
T	a forest, or a rooted tree in the architecture of Section 19	the 13-vertex gadget $\mathcal{R}(L, W)$
$E, E(G)$	an edge set	the gadget $\mathcal{R}(L)$
$H, K, J(G, H)$	a graph; a kernel multigraph; the leaf-buffered join of Lemma 12.2	the hub gadget H ; the gadget $K = \mathcal{R}(L, J)$; the gadget $J = \mathcal{R}(E)$
$c(G); c$	number of components; a theta-path length	the corner ratio $c = B_3/P_3$
$r, r_j; r$	a root multiplicity; a vector in the proof of Theorem 16.1	the hub ratio $r = B_H/P_H$, and r_m in (7.1)
s, t	vectors in the proof of Theorem 16.1; t also the leaf of Lemma 12.2	s : a gadget polynomial in (5.5); $t = 1 + x$ in Proposition 4.1
$\mathcal{P}$	the seven-element set of binary vectors in the proof of Theorem 11.1	$\mathcal{P} = P_1 P_2 P_3 P_H$
$\beta(G)$	$ E - V + c(G)$	the cyclomatic number $ E - V + 1$; equal for connected graphs
(A, B, C)	the rooted states of Lemma 3.2 at $x = -1$, with forced and waived conditions added	the same states as polynomials in x
$\mathcal{A}_b$	cumulative absolute spectrum of $ D(G, -1) $	not used

$D(G, x)$ is the domination polynomial (1.1) in both parts (Part I writes it with the macro D , the same letter).

9.3 The question from Part I

The starting point is Part I of this report, which the manuscript read at the commit named in Section 9.1. Part I specifies a connected planar graph G_{30} with 30 vertices, 31 edges, and a simple domination root -4 . It improves the order of a 33-vertex construction in Alikhani and Griswold [1]; it explicitly leaves the existence of roots -6 and -8 and the true minimum order for a root -4 unsettled.

Our first direction was to extend Part I's exact rooted-tree search from -4 to -6 . Increasing a catalogue did not produce an example. Rather than treat that failure as evidence of nonexistence, we identified an invariant that explains it: the search architecture has a built-in arithmetic obstruction. Part II develops that observation into a general theory of cycle budgets, then determines exactly the invariant's possible values through cycle rank three.

Part I already records that cycle rank two is the minimum possible for a graph with root -4 , using its construction and Oboudi's exclusion for trees and unicyclic graphs (equation (2.2), [2]). That exact minimum is *not* claimed again as a new result here.

9.4 What is proved, and what is not

The main conventional proofs are the constrained-forest lemma, the sharp 3^β bound, the root-multiplicity budget, the tree-decoration reduction, the period-four transfer identity, the complete two-cycle spectrum, and the obstruction to Part I's search architecture. The three-cycle spectrum is a computer-assisted theorem

with a fully specified finite reduction and a short, reproducible exact certificate. Its scope does not come from checking graphs up to a bounded order: it comes from the reduction of *all* subdivisions and tree decorations to finitely many cases.

No theorem in this delivery is Lean-formalized. The Python checks are not proof-assistant verification, and the manuscript has not been peer reviewed. The statements are developed independently here and are not present in Part I. A targeted literature check does not establish publication priority; in particular, the broader literature on evaluations at -1 and graph-polynomial splitting formulas must be considered before any submission [4, 5].

The existence of a graph with root -6 or -8 remains a separate question. The results below narrow the search and rule out entire structural classes; they do not substitute a congruence condition for a root construction.

10 Definitions and the elementary arithmetic obstruction

All graphs in the conclusions are finite, undirected, and simple. The multigraphs used later are only compressed descriptions of simple graphs. For a graph $G = (V, E)$, write $n = |V|$, $m = |E|$, and $c(G)$ for the number of connected components. The empty graph has $c(G) = 0$. Define

$$\beta(G) = m - n + c(G).$$

Equivalently, $\beta(G)$ is the number of edges outside any spanning forest. It is additive over components and unchanged by attaching a tree at one vertex or by subdividing an edge.

A set $S \subseteq V$ is dominating if every vertex is in S or has a neighbor in S . With $N[S]$ denoting its closed neighborhood, the domination polynomial is

$$D(G, x) = \sum_{S \subseteq V: N[S]=V} x^{|S|}. \quad (10.1)$$

We put $D(\emptyset, x) = 1$. Domination polynomials multiply over connected components. For a nonempty graph the polynomial is monic, has integer coefficients, has no positive real root, and has zero as a root. These standard facts also follow immediately from the definition [3].

Lemma 10.1 (Oddness). *For every finite simple graph, $D(G, -1)$ is an odd, nonzero integer.*

Proof. Consider the ordered pairs (A, B) of vertex sets such that $B \cap N[A] = \emptyset$. For fixed A the number of choices for B is $2^{n-|N[A]|}$. Modulo two, therefore, the number of pairs equals the number of dominating sets.

Because adjacency is symmetric, interchanging A and B is an involution on these pairs. A fixed point has $A = B$ and $A \cap N[A] = \emptyset$; since $A \subseteq N[A]$, this forces $A = B = \emptyset$. There is exactly one fixed point, so the number of dominating sets is odd. Finally, $D(G, -1) \equiv D(G, 1) \pmod{2}$. $\square$

The oddness statement is classical; it is attributed to Brouwer in the literature on $D(G, -1)$ [4]. The proof above is included to keep the later deductions independent of an external parity theorem.

Lemma 10.2 (Evaluation divisibility). *Let $k \geq 2$ be an integer. If $(x + k)^r$ divides $D(G, x)$, then*

$$(k - 1)^r \mid D(G, -1).$$

More generally, for distinct integers $k_1, \dots, k_s \geq 2$, with corresponding root multiplicities $r_1, \dots, r_s$, the product $\prod_j (k_j - 1)^{r_j}$ divides $D(G, -1)$.

Proof. Division by a monic linear polynomial with integer root preserves integral coefficients. Consequently $D(G, x) = \prod_j (x + k_j)^{r_j} Q(x)$ for $Q \in \mathbb{Z}[x]$. Evaluate at $x = -1$. $\square$

In particular a nonzero rational domination root must be a negative even integer: the rational-root theorem gives integrality, positivity is impossible, -1 is excluded by Lemma 10.1, and an odd $k \geq 3$ would make the even integer $k - 1$ divide the odd integer $D(G, -1)$. The factors corresponding to roots 0 and -2 are invisible in absolute value at $x = -1$; this is an essential limitation of the method.

11 A finite signed-state lemma for constrained forests

11.1 The constrained counting problem

Let T be a forest. At each vertex impose any combination of the following conditions: the vertex is forced selected, forced absent, or unrestricted; and its domination requirement is either retained or waived. A waived requirement models domination already supplied from outside the forest. The alternating count is

$$Z(T) = \sum_{S \text{ satisfying all conditions}} (-1)^{|S|}. \quad (11.1)$$

Conflicting selection constraints give $Z(T) = 0$.

Root each tree component. For a rooted subtree, use three states (A, B, C) . State A counts configurations with the root selected. State B counts configurations with the root absent and its requirement satisfied. State C counts configurations with the root absent and its requirement not yet satisfied, with satisfaction still possible from the parent. All other requirements in the subtree must already be satisfied. The weights are $(-1)^{|S|}$. A waived root requirement is always satisfied, so its C state is zero.

Before imposing the root's own constraints, child states (A_i, B_i, C_i) give

$$A = - \prod_i (A_i + B_i + C_i), \quad B = \prod_i (A_i + B_i) - \prod_i B_i, \quad C = \prod_i B_i. \quad (11.2)$$

For a selected root, each child may be in any state. For an absent root, every child must have its requirement satisfied internally, and at least one child must be selected to satisfy an ordinary root requirement. This explains the subtraction in B .

Waiving the root requirement replaces (A, B, C) by $(A, B + C, 0)$. Forcing selection replaces it by $(A, 0, 0)$; forcing absence replaces it by $(0, B, C)$. A complete component contributes $A + B$ at its root.

Theorem 11.1 (Constrained-forest cancellation). *For every finite forest and every set of conditions just described,*

$$Z(T) \in \{-1, 0, 1\}.$$

More precisely, every rooted signature lies in the following thirteen-element set:

$$\mathcal{S} = \{0\} \cup \{\pm e_1, \pm e_2, \pm e_3\} \cup \{\pm(e_1 - e_2), \pm(e_1 - e_3), \pm(e_2 - e_3)\}. \quad (11.3)$$

Proof. For a signature (a, b, c) introduce transformed coordinates

$$\Phi(a, b, c) = (a + b + c, a + b, b).$$

The image of $\mathcal{S}$ is $\mathcal{P} \cup (-\mathcal{P})$, where

$$\mathcal{P} = \{000, 001, 010, 011, 100, 110, 111\}.$$

Here a three-digit string denotes a vector. Notice that 101 is the only binary vector omitted. The set $\mathcal{P}$ is closed under coordinatewise multiplication: if a product has first and third coordinates one, each factor has

first and third coordinates one; each such factor must be 111, so the product cannot be 101. The signed set is therefore closed under arbitrary coordinatewise products, including the empty product 111.

Let $(p, q, r) = \prod_i \Phi(A_i, B_i, C_i)$. The recurrence produces $(-p, q - r, r)$. For the seven unsigned possibilities, these are respectively

(p, q, r)	000	001	010	011	100	110	111
$(-p, q - r, r)$	0	$(0, -1, 1)$	e_2	e_3	$-e_1$	$-e_1 + e_2$	$-e_1 + e_3$.

They all lie in $\mathcal{S}$, and the same is true after a global sign change. The three root-constraint operations also preserve $\mathcal{S}$, as follows directly from the displayed list. This proves the signature assertion by induction, without any restriction on the number of children.

For each signature in $\mathcal{S}$, $A + B \in \{-1, 0, 1\}$. Multiplication over the forest's components proves the result, including the empty forest. $\square$

The waived-requirement and pinning clauses are important. A bound proved only for ordinary trees would not justify the feedback-edge argument below: removing an edge changes both the allowed selections and which vertices are already dominated.

12 A sharp cycle-rank bound

Theorem 12.1 (Sharp alternating-sum budget). *Every finite simple graph satisfies*

$$1 \leq |D(G, -1)| \leq 3^{\beta(G)}. \quad (12.1)$$

For every $b \geq 1$ equality at the upper end is attained by a connected planar graph of maximum degree at most three with cycle rank b .

Proof of the bound. Choose a spanning forest T and orient each of the $b = \beta(G)$ remaining edges arbitrarily, solely to name its endpoints u, v . Partition the possible selections at this edge into three disjoint cases:

case	selection constraints	requirement waived after removing uv
0	u selected, v unrestricted	v
1	u absent, v selected	u
2	u, v both absent	none.

(12.2)

In case 0 the selected vertex u already dominates itself and supplies external domination to v ; thus it is unnecessary to split further according to whether v is selected. The other two cases are equally direct.

A choice of cases for all feedback edges yields a constrained-forest problem on T . Contradictory pins contribute zero. Otherwise the forest requirements, with the stated waivers, are exactly equivalent to domination in the original graph for that case. Every dominating set belongs to one and only one case vector. Therefore

$$D(G, -1) = \sum_{\eta \in \{0,1,2\}^b} Z_\eta(T).$$

Every summand belongs to $\{-1, 0, 1\}$ by Theorem 11.1, proving the upper bound. Lemma 10.1 proves the lower bound. $\square$

Lemma 12.2 (A product-preserving connection). *Let G, H be disjoint nonempty graphs with distinguished vertices u, v . Add new vertices z, t and edges uz, zv, zt . For the resulting connected join, denoted $J(G, H)$, one has*

$$D(J(G, H), -1) = -D(G, -1)D(H, -1), \quad \beta(J(G, H)) = \beta(G) + \beta(H) \quad (12.3)$$

when G, H are connected.

Proof. If z is selected, the leaf t may be selected or absent. Its two weights sum to zero, and its choice affects no other vertex; all such configurations cancel. If z is absent, t must be selected and contributes -1 . Vertex z is then already dominated, while it provides no domination to u or v . The remaining choices are independent dominating sets of G and H . The cycle-rank identity follows by counting the two added vertices and three added edges. $\square$

Proof of sharpness in Theorem 12.1. A four-cycle has $D(C_4, x) = x^4 + 4x^3 + 6x^2$ and $D(C_4, -1) = 3$. Take b disjoint four-cycles and join consecutive cycles by the operation of Lemma 12.2. On an internal four-cycle use distinct vertices for its left and right connections. The result is a planar chain of cycles with bridges and pendant leaves; all degrees are at most three. It has $6b - 2$ vertices and $7b - 3$ edges, hence cycle rank b , and

$$D(G_b^{\text{ch}}, -1) = (-1)^{b-1} 3^b.$$

For $b = 0$, a single vertex supplies the bound $|D(-1)| = 1$. $\square$

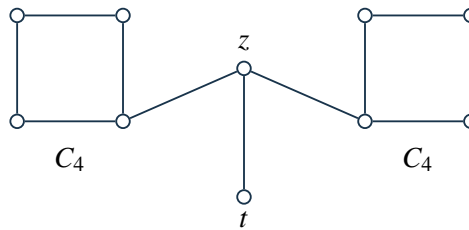


Figure 2: A connected graph with cycle rank two and $D(-1) = -9$. The selected leaf cancels all contributions with z selected; after cancellation the two four-cycles contribute independently. This also explains why retaining only a connected bare core would miss the value nine.

12.1 An exact parameterized algorithm

The proof computes $D(G, -1)$ using at most 3^β constrained-forest passes. Each pass has linear many local state operations, with states in a fixed thirteen-element set. Thus the algorithm uses $O(3^\beta(n + m))$ state operations and integer accumulations, and linear working memory apart from the counters. The output and counters require $O(\beta + 1)$ bits. This is an arithmetic-operation bound, not a claim that arbitrarily large integers have unit bit cost.

The delivered implementation records how many case vectors contribute $-1, 0, 1$. These three counts certify the resulting evaluation once the forest recursion and the partition have been checked. The algorithm's interest here is not a claim of the first fixed-parameter graph-polynomial algorithm, but the particularly strong cancellation at -1 and its arithmetic consequences.

13 Integer-root and multiplicity budgets

Theorem 13.1 (Simultaneous integer-root budget). *Suppose the distinct integer roots $-k_1, \dots, -k_s$, with $k_j \geq 2$, have multiplicities r_j . Then*

$$\prod_{j=1}^s (k_j - 1)^{r_j} \mid D(G, -1), \quad \prod_{j=1}^s (k_j - 1)^{r_j} \leq 3^{\beta(G)}. \quad (13.1)$$

In particular,

$$k \leq 3^{\beta(G)} + 1, \quad r \log_3(k - 1) \leq \beta(G) \quad (13.2)$$

for every root $-k$ of multiplicity r with $k \geq 2$.

Proof. Combine Lemma 10.2 with Theorem 12.1 and use $D(G, -1) \neq 0$. Taking logarithms gives the second formulation. $\square$

For example, a root -4 of multiplicity r requires $\beta \geq r$. A graph having both roots -4 and -8 requires $3 \cdot 7 \leq 3^\beta$, hence at least three independent cycles. A root -6 of multiplicity two requires $25 \leq 3^\beta$, so the universal bound gives $\beta \geq 3$; the exact spectrum below improves this to $\beta \geq 4$ because no permitted three-cycle value is divisible by 25.

Remark 13.2 (What is sharp). The 3^β evaluation bound is sharp. This does *not* imply that $-(3^\beta + 1)$ is a domination root of an extremal graph. Nor does it imply that the cycle lower bound for every prescribed root multiplicity is attained. Divisibility of an evaluation is necessary, never sufficient.

The case $k = 2$ has $\log_3(k - 1) = 0$, so there is no restriction on the multiplicity of -2 from this invariant. This agrees with ordinary forests: the disjoint union of r edges has domination polynomial $x^r(x + 2)^r$ and cycle rank zero.

14 Pruning tree decorations without changing the invariant

The constrained-forest theorem has a much stronger specialization when there are no constraints.

Lemma 14.1 (Ordinary tree signatures). *For a rooted tree at $x = -1$ its three-state signature is exactly one of*

$$\pm(-1, 0, 1), \quad \pm(0, 1, 0). \quad (14.1)$$

In particular, $D(T, -1) = \pm 1$ for every tree.

Proof. A leaf has signature $(-1, 0, 1)$. Up to a global sign, the transformed coordinates of the two displayed types are $(0, -1, 0)$ and $(1, 1, 1)$. If at least one child is of the first type, the product coordinates have first and third entries zero and second entry ± 1 , producing $(0, \pm 1, 0)$. Otherwise they equal $\pm(1, 1, 1)$, producing $\pm(-1, 0, 1)$. Induction proves the claim. $\square$

Theorem 14.2 (Tree-decoration deletion). *Let H be a graph and attach a rooted tree T_v at each $v \in V(H)$ by identifying its root with v , with the trees otherwise disjoint. Let the resulting graph be G . Write the signature of T_v as either $\epsilon_v(-1, 0, 1)$ or $\epsilon_v(0, 1, 0)$, where $\epsilon_v \in \{-1, 1\}$, and let U be the vertices of the first type. Then*

$$D(G, -1) = \left(\prod_{v \in V(H)} \epsilon_v \right) D(H[U], -1). \quad (14.2)$$

Proof. An ordinary vertex of H , before interacting with its neighbors in H , has signature $(-1, 0, 1)$. The first type is exactly this signature times ϵ_v . The second type has root absent, already dominated, and never selected. It therefore neither needs domination from nor supplies domination to its neighbors; it can be deleted, leaving the multiplicative factor ϵ_v . Applying this observation at every vertex proves the identity. $\square$

For a graph with nonempty 2-core, all vertices outside that core form rooted tree decorations of its vertices, together with separate tree components. Apply Theorem 14.2, and then repeat if vertex deletions create new tree decorations. The process terminates, and proves that $|D(G, -1)|$ equals the absolute evaluation of a graph whose nonempty components have minimum degree at least two; separate forest components only supply signs. At every step cycle rank does not increase. Indeed, the cycle space of a subgraph embeds in that of the original graph.

The reduced graph need not stay connected, even when G is connected. This is why products of cycle values enter the exact spectra. The example in the preceding figure reduces to two disjoint four-cycles.

[Added 30 September 2026, batch 67] Relation to Part I's Certificate B. Part I's Certificate B (Section 5, equation (5.6)) also strips pendant trees down to the 2-core, but it works for every x : each stripped tree is absorbed into its core vertex through the three products T_v, U_v, V_v , and the core is then enumerated. Theorem 14.2 is the specialization to $x = -1$. There the absorbed data take only the four values (14.1), so a vertex of the second type can be deleted outright and the evaluation reduces to a smaller graph. At $x = -1$ the two compute the same value by different routes; the deletion is what Part II adds. Neither source states this comparison; it is the present write's.

Corollary 14.3 (Sign reversal without adding a cycle). *Attaching a pendant path with two new vertices at any existing vertex multiplies $D(G, -1)$ by -1 and leaves cycle rank unchanged.*

Proof. The rooted three-vertex path, rooted at an end, has signature $(1, 0, -1) = -(-1, 0, 1)$; apply Theorem 14.2. $\square$

15 A period-four transfer identity for all subdivisions

15.1 A three-state expansion

Use states 0, 1, 2 at each graph vertex: respectively an unselected unmarked vertex, a selected vertex, and an unselected vertex marked by inclusion–exclusion. Put

$$w = (1, -1, -1), \quad W = \text{diag}(1, -1, -1), \quad M_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \quad (15.1)$$

Then

$$D(G, -1) = \sum_{\sigma: V \rightarrow \{0,1,2\}} \prod_{v \in V} w_{\sigma(v)} \prod_{uv \in E} (M_1)_{\sigma(u), \sigma(v)}. \quad (15.2)$$

To verify this identity, first fix the selected set S . A state-2 mark is allowed exactly at an unselected vertex with no selected neighbor. Summing over the possible marks multiplies $(-1)^{|S|}$ by $(1-1)^h$, where h is the number of undominated vertices. The contribution is therefore $(-1)^{|S|}$ for a dominating set and zero otherwise.

For a path of ℓ edges, sum the internal states but do not include the endpoint weights. The resulting matrix is

$$M_\ell = M_1(WM_1)^{\ell-1}, \quad \ell \geq 1. \quad (15.3)$$

Direct multiplication gives

$$\begin{aligned} M_2 &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, & M_3 &= \begin{pmatrix} -1 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}, \\ M_4 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = W, & (WM_1)^4 &= I. \end{aligned} \quad (15.4)$$

Theorem 15.1 (Four-subdivision invariance). *Replacing any edge of a simple graph by a path with five edges leaves $D(G, -1)$ unchanged. More generally, the contribution of an internally disjoint path depends on its length only modulo four.*

Proof. Equations (15.3)–(15.4) give $M_{\ell+4} = M_\ell$. Replace the corresponding factor in the three-state expansion; all endpoint interactions with the rest of the graph are retained. $\square$

15.2 Compressed kernels

A connected graph of minimum degree at least two is either a cycle or, after suppressing its degree-two vertices, a connected multigraph K of minimum degree at least three. Multiple edges and loops are allowed in K . A loop represents a cycle returning to its branch vertex; in the simple expansion it must have length at least three. Each kernel edge $e = uv$ is assigned a path length ℓ_e .

If $b = \beta(K) \geq 2$, the kernel has

$$|V(K)| \leq 2b - 2, \quad |E(K)| = |V(K)| + b - 1 \leq 3b - 3. \quad (15.5)$$

Indeed, the degree sum is at least three times the vertex count, whereas $2|E| = 2|V| + 2b - 2$.

The exact evaluation of an expansion is

$$\mathcal{Z}_K(\ell) = \sum_{\sigma: V(K) \rightarrow \{0,1,2\}} (-1)^{|\{v: \sigma(v) \neq 0\}|} \prod_{e=uv \in E(K)} (M_{\ell_e})_{\sigma(u), \sigma(v)}. \quad (15.6)$$

For a loop use the corresponding diagonal matrix entry. Equation (15.6) is valid because each branch-vertex weight is counted once and internal path vertices are disjoint.

Every assignment of residues in $\{1, 2, 3, 4\}$ is realizable by a simple expansion. Replace a loop of length one or two by length five or six. For parallel edges with residue one, keep at most one direct edge and use length five for the others. Distinct paths receive disjoint internal vertices. These changes preserve all residues and the exact evaluation. Thus a residue enumeration gives the exact image, not merely an upper bound containing unrealizable cases.

Cycles themselves satisfy

$$D(C_q, -1) = \text{tr}((WM_1)^q) = \begin{cases} 3, & q \equiv 0 \pmod{4}, \\ -1, & q \not\equiv 0 \pmod{4}, \end{cases} \quad q \geq 3. \quad (15.7)$$

16 The exact two-cycle spectrum: a conventional proof

Define the cumulative absolute spectrum

$$\mathcal{A}_b = \{|D(G, -1)| : G \text{ finite simple}, \beta(G) \leq b\}.$$

Theorem 16.1 (Bicyclic spectrum). *One has*

$$\mathcal{A}_0 = \{1\}, \quad \mathcal{A}_1 = \{1, 3\}, \quad \mathcal{A}_2 = \{1, 3, 7, 9\}. \quad (16.1)$$

Every value in these sets is realized by a connected graph within the stated cycle budget, and either sign can be realized.

Proof. Forests contribute absolute value one. By pruning, a graph with cycle rank at most one reduces to cycles and forests, with at most one cycle. This gives the first two assertions by (15.7).

For a connected minimum-degree-two component of cycle rank two, the kernel bound gives at most two branch vertices. The possible kernels are exactly a figure-eight, a dumbbell, and a theta graph: respectively two loops at one vertex; one loop at each endpoint of a connecting edge; and three parallel edges. We compute their exact ranges next.

For a figure-eight of cycle lengths a, b , write $d_a = \text{diag}(M_a)$ as a column vector. Then its evaluation is $d_a^T W d_b$. The distinct possible vectors d_a are

$$t = (1, 1, 1)^T, \quad s = (-1, 0, 0)^T, \quad r = (1, -1, -1)^T.$$

Their pairings give only $-1, 1, 3$.

For a dumbbell with cycle lengths a, b and connecting-path length ℓ , its evaluation is $(Wd_a)^T M_\ell (Wd_b)$. The possible vectors Wd_a are r, s, t . In this order, the complete pairing matrices for $\ell \equiv 1, 2, 3, 0 \pmod{4}$ are

$$\begin{pmatrix} -1 & 1 & -1 \\ 1 & 1 & -3 \\ -1 & -3 & 7 \end{pmatrix}, \begin{pmatrix} 1 & 1 & -3 \\ 1 & -1 & 1 \\ -3 & 1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & -1 & 3 \\ 1 & 3 & -7 \end{pmatrix}, \begin{pmatrix} -1 & -1 & 3 \\ -1 & 1 & -1 \\ 3 & -1 & -1 \end{pmatrix}. \quad (16.2)$$

Their entries have absolute values $1, 3, 7$.

For a theta graph with path lengths a, b, c , write

$$M_j = \begin{pmatrix} p_j & q_j & q_j \\ q_j & r_j & s_j \\ q_j & s_j & r_j \end{pmatrix}.$$

Its evaluation is

$$p_a p_b p_c - 4q_a q_b q_c + 2r_a r_b r_c + 2s_a s_b s_c. \quad (16.3)$$

The quadruples (p_j, q_j, r_j, s_j) for residues $j = 1, 2, 3, 4$ are

$$(1, 1, 1, 0), \quad (-1, 0, 0, 1), \quad (-1, -1, 0, -1), \quad (1, 0, -1, 0).$$

The q product is nonzero only when all residues lie in $\{1, 3\}$; the r product only for $\{1, 4\}$; and the s product only for $\{2, 3\}$. Two products can be nonzero together only when all residues are one or all are three. In these two cases the values are -1 and 1 . If only the q product is nonzero, the value is ± 3 ; if only the r or s product is nonzero, the value is in $\{-3, -1, 1, 3\}$; and if none is nonzero, the value is ± 1 . Thus theta graphs add no other absolute value.

Finally, a disconnected reduced graph of total cycle rank two may have two cycle components, adding the product $3 \cdot 3 = 9$ and no other value. Tree pruning proves that this exhausts arbitrary graphs, not just bare cores. Values 1 and 3 are supplied by a vertex and C_4 . Two copies of C_4 joined by a direct edge have value 7 , by the first matrix in (16.2); their leaf-buffered join has value -9 . Corollary 14.3 supplies the opposite signs without adding cycles. $\square$

Corollary 16.2 (A two-cycle root exclusion). *If $\beta(G) \leq 2$, every nonzero rational domination root lies in $\{-2, -4, -8, -10\}$.*

In particular $D(G, -6) \neq 0$ for every such graph.

Proof. A root $-k$ requires $k - 1$ to divide one of $1, 3, 7, 9$. Their positive divisors are exactly $1, 3, 7, 9$. $\square$

The candidate set is not an attainment statement: this proof does not construct a root -8 or -10 . It does show that searching for a root -6 within any family of cycle rank two is futile, even with arbitrary numbers of leaves, arbitrary tree shapes, and arbitrarily long subdivisions. The obstruction does not depend on the special zero- C condition discussed later.

17 The exact three-cycle spectrum and its finite certificate

Theorem 17.1 (Tricyclic spectrum; exact finite certificate). *For graphs of cycle rank at most three,*

$$\mathcal{A}_3 = \{1, 3, 5, 7, 9, 11, 15, 17, 21, 27\}. \quad (17.1)$$

Each value occurs in a connected graph, and either sign occurs without increasing its cycle rank.

17.1 Exhaustion of kernels

A minimum-degree-three connected multigraph with cycle rank three has $1 \leq v \leq 4$ vertices and $e = v + 2$ edges. The following finite enumeration is exhaustive. For each v , list multisets of e unordered endpoint pairs from $\{0, \dots, v-1\}$, allowing repeated pairs and loops. Keep exactly the connected ones in which every degree, with loops counted twice, is at least three. Identify two multisets only if a vertex permutation maps one to the other. This yields respectively 1, 4, 5, 5 kernels on 1, 2, 3, 4 vertices, hence fifteen in total. For each kernel evaluate (15.6) for every residue assignment. The number of evaluations is

$$1 \cdot 4^3 + 4 \cdot 4^4 + 5 \cdot 4^5 + 5 \cdot 4^6 = 26\,688.$$

No hypothesis about bounded graph order is involved: a path may be arbitrarily long and still has one of the four transfer matrices.

ID	v	Edge multiset	Absolute image of $\mathcal{Z}_K$
0	1	00, 00, 00	$\{1, 3\}$
1	2	00, 00, 01, 11	$\{1, 3, 7\}$
2	2	00, 01, 01, 01	$\{1, 3\}$
3	2	00, 01, 01, 11	$\{1, 3, 5, 7\}$
4	2	01, 01, 01, 01	$\{1, 3\}$
5	3	00, 01, 01, 12, 22	$\{1, 3, 5, 7\}$
6	3	00, 01, 02, 11, 22	$\{1, 3, 7, 9, 17\}$
7	3	00, 01, 02, 12, 12	$\{1, 3, 5, 7\}$
8	3	00, 01, 12, 12, 12	$\{1, 3, 7\}$
9	3	01, 01, 02, 02, 12	$\{1, 3\}$
10	4	00, 01, 12, 12, 23, 33	$\{1, 3, 5, 7, 9, 15, 17\}$
11	4	00, 01, 12, 13, 22, 33	$\{1, 3, 7, 9, 11, 17\}$
12	4	00, 01, 12, 13, 23, 23	$\{1, 3, 5, 7\}$
13	4	01, 01, 02, 13, 23, 23	$\{1, 3, 5, 7\}$
14	4	01, 02, 03, 12, 13, 23	$\{1, 3, 5\}$

Table 3: The complete three-cycle kernel certificate. Vertices are numbered from zero; ij is an unordered edge, repetitions are parallel edges, and ii is a loop. Every length-residue assignment is realized by a simple subdivision. IDs agree with the JSON certificate and verification program.

17.2 Proof from the certificate

Proof of Theorem 17.1. Apply tree pruning. A remaining component is a cycle, a two-cycle kernel expansion, or a three-cycle kernel expansion. Table 3 gives the exact absolute image of the last type by the enumeration above and formula (15.6). The union of its rows is

$$\{1, 3, 5, 7, 9, 11, 15, 17\}.$$

Components of cycle ranks two and one may contribute a product. The only additional value is $7 \cdot 3 = 21$; three cycle components additionally supply $3^3 = 27$. The value 9 was already present. These are all partitions of a total cycle budget at most three, so no other absolute evaluation is possible.

Every residue assignment in the table is realizable by a simple expansion, as proved after (15.6). Explicit witnesses for every value are listed in Table 4 and the accompanying JSON data. Products are made connected by Lemma 12.2, which preserves the additive cycle budget. Sign reversal follows from Corollary 14.3. $\square$

The finite part of this proof is the exact arithmetic table, not a statistical inference. The program independently regenerates the kernels rather than reading an asserted catalogue. It tests every vertex

permutation on at most four vertices, so no graph-isomorphism package is trusted. The proof still has the usual limitations of a computer-assisted argument: its conventional reductions and executable certificate require review, and no proof assistant has checked their correspondence.

Corollary 17.2 (Three-cycle rational-root restrictions). *If $\beta(G) \leq 3$, every nonzero rational domination root is among*

$$-2, -4, -6, -8, -10, -12, -16, -18, -22, -28.$$

Thus roots $-14, -20, -24, -26$ each require at least four independent cycles. A root -6 of multiplicity at least two also requires at least four.

Proof. List the positive divisors of the integers in (17.1), then add one and negate. None of those integers is divisible by 25, which proves the multiplicity assertion. $\square$

17.3 The congruence barrier is already sharp at three cycles

Let X_5 have vertex set $\{0, \dots, 7\}$ and edge set

$$\{03, 05, 01, 02, 12, 16, 27, 34, 45, 67\}, \quad (17.2)$$

where ij denotes the unordered pair $\{i, j\}$. It has eight vertices, ten edges, cycle rank three, and $D(X_5, -1) = 5$. In Table 3 it is kernel 7 with lengths $(4, 1, 1, 1, 3)$. Substitution in (15.6) proves the evaluation, and direct enumeration of its 256 vertex subsets independently confirms it.

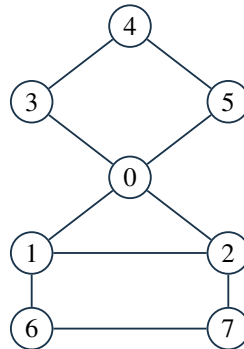


Figure 3: An eight-vertex tricyclic graph with $D(X_5, -1) = 5$. The congruence required by a prospective root -6 is therefore compatible with cycle rank three. This graph is not asserted to have that root.

The independently enumerated polynomial is

$$D(X_5, x) = x^3(11 + 46x + 51x^2 + 28x^3 + 8x^4 + x^5),$$

so $D(X_5, -6) = 407\,160 \neq 0$. Consequently the lower bound of three cycles for a root -6 is optimal for the obstruction using divisibility of $D(-1)$ alone; it is not yet an existence result at that cycle rank.

18 Exact cycle complexity of small alternating evaluations

The same classification resolves a different extremal question outright. For a positive odd integer a , define

$$\mu(a) = \min\{\beta(G) : G \text{ connected and } |D(G, -1)| = a\}.$$

Existence of connected realizations for every odd evaluation was established by Alikhani [4]; here only the explicit witnesses below are needed. Theorem 12.1 gives $\mu(a) \geq \lceil \log_3 a \rceil$.

Corollary 18.1 (Exact minima through cycle budget three). *The positive odd integers with $\mu(a) \leq 3$ are exactly those in (17.1), and their minima are those in Table 4. In particular,*

$$\mu(5) = 3, \quad \mu(7) = \mu(9) = 2, \quad \mu(3^b) = b \quad (b \geq 0).$$

For $a = 13, 19, 23, 25$, one has $\mu(a) \geq 4$.

Proof. Compare the cumulative spectra $\mathcal{A}_0, \dots, \mathcal{A}_3$ and use the connected witnesses. For powers of three, the universal lower bound and the sharpness construction match for every b . $\square$

a	$\mu(a)$	Witness order	Signed value	Construction
1	0	1	-1	A single vertex
3	1	4	3	C_4
5	3	8	5	Kernel 7, lengths (4, 1, 1, 1, 3)
7	2	8	7	Two C_4 cycles joined by an edge
9	2	10	-9	Leaf-buffered product connection
11	3	13	11	Kernel 11, lengths (4, 1, 1, 1, 4, 4)
15	3	12	15	Kernel 10, lengths (4, 1, 1, 3, 1, 4)
17	3	12	17	Kernel 6, lengths (4, 1, 1, 4, 4)
21	3	14	-21	Leaf-buffered product connection
27	3	16	27	Leaf-buffered product connection

Table 4: Exact minimum cycle ranks and explicit connected witnesses. The stated orders are not asserted to be minimum orders. Full edge lists are supplied in data/02-cycle-budgets-spectrum_witnesses.json (the manuscript's data/spectrum_witnesses.json).

For example, the logarithmic bound alone would allow $\mu(5) = 2$, but Theorem 16.1 rules it out. This illustrates why the exact spectrum is substantially stronger than an absolute-value bound. The vertex counts in the table are explicit witness sizes, not claims of minimum order among graphs realizing the same evaluation.

19 Why Part I's four-cycle search cannot find another root

19.1 The architecture and its zero- C condition

Part I's successful search (Section 7) starts with a four-cycle. It adds a vertex h adjacent to two consecutive cycle vertices and gives h at least one pendant leaf. It then attaches rooted trees by a single edge from their roots to specified cycle vertices. At the proposed root $x = -k$, the admitted trees satisfy

$$C_T(-k) = 0, \tag{19.1}$$

where $C_T(x)$ is the generating polynomial for selections that dominate all vertices of T except its root, which is absent and undominated. The search may combine several such trees into forests at each site. Its original four-cycle has one undecorated corner, but the theorem below does not require that restriction.

Theorem 19.1 (Zero- C four-cycle obstruction). *Let $k \geq 3$ be an integer and construct G by the architecture just described, with any finite number of attached rooted trees satisfying (19.1). Then*

$$|D(G, -1)| = 3.$$

If $D(G, -k) = 0$, necessarily $k = 4$, and that root is simple.

Proof. The polynomial C_T has integer coefficients, so

$$C_T(-1) \equiv C_T(-k) = 0 \pmod{k-1}.$$

By Lemma 14.1, $C_T(-1) \in \{-1, 0, 1\}$. Since $k-1 \geq 2$, we must have $C_T(-1) = 0$, so the signature of T is $\epsilon(0, 1, 0)$. Such an attached tree has root absent and already dominated; it contributes only its sign, and neither requires nor supplies domination across the attachment edge. Removing all these trees thus changes $D(-1)$ by a sign only.

At the hub h , the presence of one or more leaves gives rooted-star signature $(0, (-1)^q, 0)$, where $q \geq 1$ is the number of leaves. Therefore h is absent and already dominated after cancellation; delete it, again retaining only a sign. What remains is precisely C_4 , with evaluation three. Thus $|D(G, -1)| = 3$.

A root $-k$ now requires $k-1 \mid 3$. Because $k \geq 3$, the only possibility is $k = 4$. A multiple root -4 would require $9 \mid D(G, -1)$, which is impossible. Hence the root is simple. $\square$

The zero- C qualification is part of the theorem. It is not legitimate to conclude that every graph obtained by arbitrary decorations of the same core has $|D(-1)| = 3$. Without that qualification the broader bicyclic theorem still excludes -6 , but it allows the necessary congruences for -8 and -10 .

19.2 The 30-vertex example revisited

For the exact edge list (2.1) of Part I, the feedback-edge algorithm gives

$$D(G_{30}, -1) = -3.$$

Its nine case vectors contribute three copies of -1 and six zeros. Consequently the already established root -4 is automatically simple: there is no need to recompute the derivative or its large residual factor for this particular conclusion. The existence of the root itself remains Part I's polynomial certificate (Section 5), not a consequence of the value -3 .

A prospective search for -6 must therefore leave cycle rank two altogether. A search for -8 can in principle retain two cycles, but it must escape the zero- C four-cycle architecture or otherwise change the invariant from absolute value three to seven. This is a precise distinction between an obstruction to a *graph class* and an obstruction to a *parameterized search design*.

20 Verification package, reproducibility, and limitations

20.1 What the executable certificate checks

The file `code/verify.py` (shipped here as `code/02-cycle-budgets-verify.py`; see the note in Section 20.2) uses Python's standard library and exact integers. It regenerates the kernel catalogue, transfer checks, witness graphs, and output tables without any downloaded data. The recorded run has the following scope.

Check	Cases
All labeled simple graphs on zero through five vertices	1,100
Additional constrained forests, fixed random seed	300
Additional graphs, fixed random seed	120
Direct selected-bit checks of path matrices	lengths 1–12
Bicyclic kernels and residue assignments	3 / 144
Tricyclic kernels and residue assignments	15 / 26,688
Independent small expanded-kernel brute-force checks	78
Sharpness construction evaluations	cycle ranks 1–6

For every small graph, direct vertex-subset enumeration of the full polynomial is compared with the feedback-edge algorithm at -1 . The tests also verify oddness, the 3^B bound, the low-cycle spectra where applicable, and the divisibility implication for tested integer roots. The program checks arbitrary-arity signature closure by verifying closure of the transformed state set under coordinatewise multiplication; it does not infer that closure from a bound on tree size.

The transfer matrices are independently checked by enumerating selected internal vertices of short paths, rather than by using the same matrix multiplication twice. Small kernel expansions are also evaluated directly as simple graphs. The exact witness data give edges, cycle ranks, signs, and subdivision lengths. Seeded tests are supplementary implementation checks, not the proof of a universal assertion.

20.2 Reproduction

From the extracted directory run

```
python3 code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The verification script writes `results/verification.json`, `data/kernel_certificate.json`, and `data/spectrum_witnesses.json`. The compact tables are embedded in the self-contained TeX source, so the PDF can be rebuilt without rerunning the computation. The supplementary code/`check_repo_example.py` checks Part I's exact 30-vertex edge list at -1 and records the nine feedback-edge contributions.

[Added 30 September 2026, batch 67] Shipped names; do not run these commands here. The commands and paths above are the manuscript's, for its delivered directory. In this report they mean something else: `code/verify.py` is Part I's verifier, and `article.tex` does not exist. Part II's files are shipped with the prefix `02-cycle-budgets-`: the verifier is `code/02-cycle-budgets-verify.py`, the example check is `code/02-cycle-budgets-check_repo_example.py`, the recorded runs are `data/02-cycle-budgets-verification.json` and `data/02-cycle-budgets-example_checks.json`, and the two certificates are `data/02-cycle-budgets-kernel_certificate.json` and `data/02-cycle-budgets-spectrum_witnesses.json`. Both programs write into the directory above their own. Run in place, the verifier would overwrite Part I's `results/verification.json` and add unprefixed files to `data/`. The example check would add `results/example_checks.json`; it also imports the verifier as the module `verify`, which the prefixed file name cannot provide. Run them only on a copy under their delivered names, as the report's README describes. The PDF of this report is built from `domination_root_30.tex`.

20.3 Trust boundaries

The sharp bound, arithmetic deductions, and two-cycle classification have full proofs in Part II. The three-cycle theorem additionally depends on the finite table. The bundled computation is exact and reproducible, but has not been translated into a proof assistant. No unrefereed theorem elsewhere in ProveIt is needed for these new structural proofs. Part I's root- -4 construction is used only as a motivation and an application to a specific known example.

The literature review confirms the relevant problem statements and standard background, but is not exhaustive. Some publisher material is accessible only as an abstract. We therefore make no claim that every lemma or the 3^B inequality is absent from all prior literature. The primary research contribution presented for review is the complete proof package, the exact low-cycle classification, the simultaneous root-budget consequences, and their application to the pinned search architecture.

21 Further research questions and concrete next steps

21.1 The first admissible cycle rank for a root -6

Question 21.1. Does there exist a connected graph G with $\beta(G) = 3$ and $D(G, -6) = 0$?

The obstruction at -1 allows this: kernel 7 already has a simple expansion with evaluation five. A focused search should begin with the kernel rows admitting values divisible by five, namely rows 3, 5, 7, 10, 12, 13, and 14 of Table 3. Values ± 5 and ± 15 are the only three-cycle evaluations compatible with the root. This restriction is necessary for the reduced graph after pruning, not a license to drop arbitrary decorations from an evaluation at -6 .

A useful approach is to work with exact rooted signatures at -6 while simultaneously maintaining their four-state signatures at -1 . The latter provide a cheap rejection certificate; the former retain the root equation. The eight-vertex X_5 is a starting core, not a solution.

21.2 The remaining bicyclic root candidates

Question 21.2. Can -8 or -10 be a domination root of a graph with cycle rank two?

Such a graph must have $D(-1) = \pm 7$ or ± 9 , respectively. For -8 , the nontrivial reduced component must come from the dumbbell range. For -10 , the terminal reduction must supply two factors of absolute value three, hence two cycle components; no connected bare bicyclic core has absolute value nine. This specifies the type of cancellation that a successful tree decoration must preserve.

21.3 Multiplicity as an independent structural problem

Question 21.3. What is the minimum cycle rank for a root -4 of multiplicity r ? In particular, can multiplicity two occur at cycle rank two?

The bound is at least r , and the case $r = 1$ has minimum two by Part I's example and the published unicyclic exclusion (equation (2.2)). For $r = 2$ and cycle rank two, the evaluation must be ± 9 ; for $r = 3$ and cycle rank three it must be ± 27 . These necessary conditions rule out the zero- C architecture completely, since it forces simplicity. Part I's arbitrary-multiplicity constructions (Theorem 6.4) can be compared with these lower bounds rather than being mistaken for optimal cycle budgets.

21.4 Higher spectra and an extremal evaluation function

Question 21.4. Determine $\mathcal{A}_4$, and characterize the function $\mu(a)$ for odd positive integers a .

The same method is finite for each fixed budget: kernels have at most $2b - 2$ vertices and $3b - 3$ edges, and every edge has four residue states. Kernel automorphisms and repeated transfer tensors can reduce enumeration. A certified calculation for $b = 4$ would settle whether the next missing odd values 13, 19, 23, 25 have cycle complexity four and would sharpen root exclusions beyond -14 .

The identity $\mu(3^b) = b$ gives one exact infinite subsequence. The value $\mu(5) = 3 > \lceil \log_3 5 \rceil$ shows that the logarithmic lower bound is not always tight. Is the excess over that bound uniformly bounded, or can arithmetic gaps force arbitrarily large excess?

21.5 Equality and stability in the cycle budget

Question 21.5. Which connected graphs attain $|D(G, -1)| = 3^{\beta(G)}$? What structural restrictions follow from being close to this maximum?

For a chosen spanning forest, equality means that all 3^β case vectors are nonzero and have the same sign. This is an exact certificate criterion, but not yet an intrinsic graph classification. A more useful answer would characterize the reduced cores and permitted decorations. The subcubic chain construction supplies extremizers at every budget; it need not exhaust them.

21.6 A practical formalization route

A Lean development can separate three layers. First formalize dominating sets, the three-way feedback-edge partition, and the thirteen-state forest invariant. Second formalize the small integer matrices and the period-four identity, so that path suppression has a precise semantic lemma. Third reflect the bounded kernel enumeration and its integer sums into certified finite computations. This would turn the finite table into a kernel-checked theorem instead of merely porting the Python code.

The first target should be the universal budget and the purely mathematical bicyclic exclusion; neither needs a large computed certificate. The exact tricyclic table is a natural subsequent reflection exercise. This is a proposed development plan, not a statement that any such formalization has already been executed.

[Added 30 September 2026, batch 67] Formal status in ProveIt. No Lean or Rocq development in ProveIt defines dominating sets or the domination polynomial, so neither part of this report has a formal counterpart. Placement of this report in the repository confers no formal status on any of its statements.

22 Conclusion of Part II

An unsuccessful numerical search became informative only after identifying what its cancellations preserve. At $x = -1$, arbitrary constrained forests collapse to a thirteen-element signed state set. Every feedback edge then costs at most a factor of three, yielding a sharp cycle budget and a simultaneous restriction on integer roots and their multiplicities.

Tree pruning and a period-four transfer identity make the invariant fully classifiable at low cycle rank. They exclude -6 from all graphs with at most two independent cycles, exclude -14 from all graphs with at most three, and determine several exact minima for alternating evaluations. They also explain why Part I's zero- C four-cycle architecture is intrinsically a simple-root- -4 mechanism, not a general route to new integer roots. These conclusions leave well-defined construction problems at the first cycle ranks where the obstruction permits them.

A Reproducing the artifact

The archive contains the $\text{\LaTeX}$ source and compiled PDF, a machine-readable edge list and coefficient table, the 118-term leaf certificate, the exact verification report, and the reproducible search program. From the archive's top directory, run

```
python3 code/verify.py
python3 code/search.py
```

Python 3.10 or later is sufficient. The first command recomputes all certificates, runs the tests, and writes `results/verification.json`. The second reconstructs the rooted catalogue from scratch and writes

results/search.json. Both commands stop with an error if their explicit verification checks fail; the tests do not rely on Python’s optionally disabled `assert` statement.

To rebuild the document using a standard TeX Live installation:

```
pdflatex -interaction=nonstopmode -halt-on-error domination_root_30.tex
pdflatex -interaction=nonstopmode -halt-on-error domination_root_30.tex
```

The graph drawing is embedded as TikZ code in this source. No external image or font file is needed. The output PDF was rendered and visually inspected before packaging.

Authorship and status. This note was produced in response to a request for an original research attempt. The construction, proof, and verification artifacts were developed in the session with ChatGPT. The note is not peer reviewed and has not been formally verified in Lean. The exact finite certificate is intended to make external checking straightforward.

[Added 30 September 2026, batch 67] **Part II.** This appendix describes Part I’s files. Part II’s programs and records carry the prefix `02-cycle-budgets-`; Section 20 describes them and the report’s README gives commands that run them on a copy. Part II, too, was prepared with ChatGPT; it is not peer reviewed and not formally verified, and its three-cycle classification is computer-assisted (Section 17).

B Compact independent formula for the kernel table

The following pseudocode is sufficient to reproduce every row once the kernel’s endpoint list is given. It uses only the four matrices printed in (15.1) and (15.4).

```
def kernel_value(vertices, edges, lengths):
    answer = 0
    for state in product(range(3), repeat=vertices):
        term = (-1) ** sum(s != 0 for s in state)
        for (u, v), length in zip(edges, lengths):
            term *= M[1 + (length - 1) % 4][state[u]][state[v]]
        answer += term
    return answer

values = {
    abs(kernel_value(vertices, edges, lengths))
    for lengths in product(range(1, 5), repeat=len(edges))
}
```

A loop $u = v$ requires no exceptional code. The interpretation of the compressed kernel is a subdivision with a genuine simple cycle, not a looped graph’s domination polynomial.

C Source and novelty audit

Source or component	Role and status
Part I, read at the commit of Section 9.1	Motivation; exact 30-vertex example; zero- C search architecture. The root construction and its prior minimum cycle rank are credited, not claimed as new.
Alikhani–Griswold [1]	Primary statement of the root- -4 counterexample and follow-up questions. Part II does not claim their counterexample as its own.

Source or component	Role and status
Alikhani [4]	Prior work on evaluations at -1 , including connected realizability of every odd value and the attribution of oddness to Brouwer.
Kotek et al. [5]	Prior general recurrence and splitting theory. The local recurrences here are reproved; graph-polynomial dynamic programming itself is not claimed as new.
Sections 11–16 and 19	New derivations relative to Part I, with conventional proofs; broader publication priority unverified.
Section 17 and dependent corollaries	Exact finite computer-assisted classification; all-graph reduction proved in text, certificate executed, not Lean-checked.

References

- [1] S. Alikhani and M. Griswold, *On the Integer Domination Root Conjecture*, arXiv:2608.00109v1, 2026. arxiv.org/abs/2608.00109v1. The selected minimality question is Section 4, question 2; the infinite-family question is question 3. Accessed 18 September 2026. [Added 30 September 2026, batch 67:] Part II inspected the HTML version, <https://arxiv.org/html/2608.00109v1>; Section 4 lists the further-root and minimum-order questions.
- [2] M. R. Oboudi, *On the roots of domination polynomial of graphs*, Discrete Applied Mathematics **205** (2016), 126–131. doi:10.1016/j.dam.2015.12.010. Theorem 7 supplies the trees/unicyclic exclusion used in (2.2); Lemma 1 is the general domination inclusion–exclusion identity. [Added 30 September 2026, batch 67:] In Part II the trees/unicyclic exclusion is used only as credited context for the previously known minimum cycle rank of a root -4 ; none of the new obstruction proofs depends on it.
- [3] S. Alikhani and Y.-H. Peng, *Introduction to Domination Polynomial of a Graph*, Ars Combinatoria **114** (2014), 257–266; arXiv:0905.2251. <https://arxiv.org/abs/0905.2251>.
- [4] S. Alikhani, *The Domination Polynomial of a Graph at -1* , Graphs and Combinatorics **29** (2013), 1175–1181. <https://doi.org/10.1007/s00373-012-1211-x>. The publisher abstract and bibliography were inspected; the proofs of Part II do not assume unseen results from the full article.
- [5] T. Kotek, J. Preen, F. Simon, P. Tittmann, and M. Trinks, *Recurrence Relations and Splitting Formulas for the Domination Polynomial*, Electronic Journal of Combinatorics **19**(3) (2012), P47. <https://www.combinatorics.org/ojs/index.php/eljc/article/view/v19i3p47>.